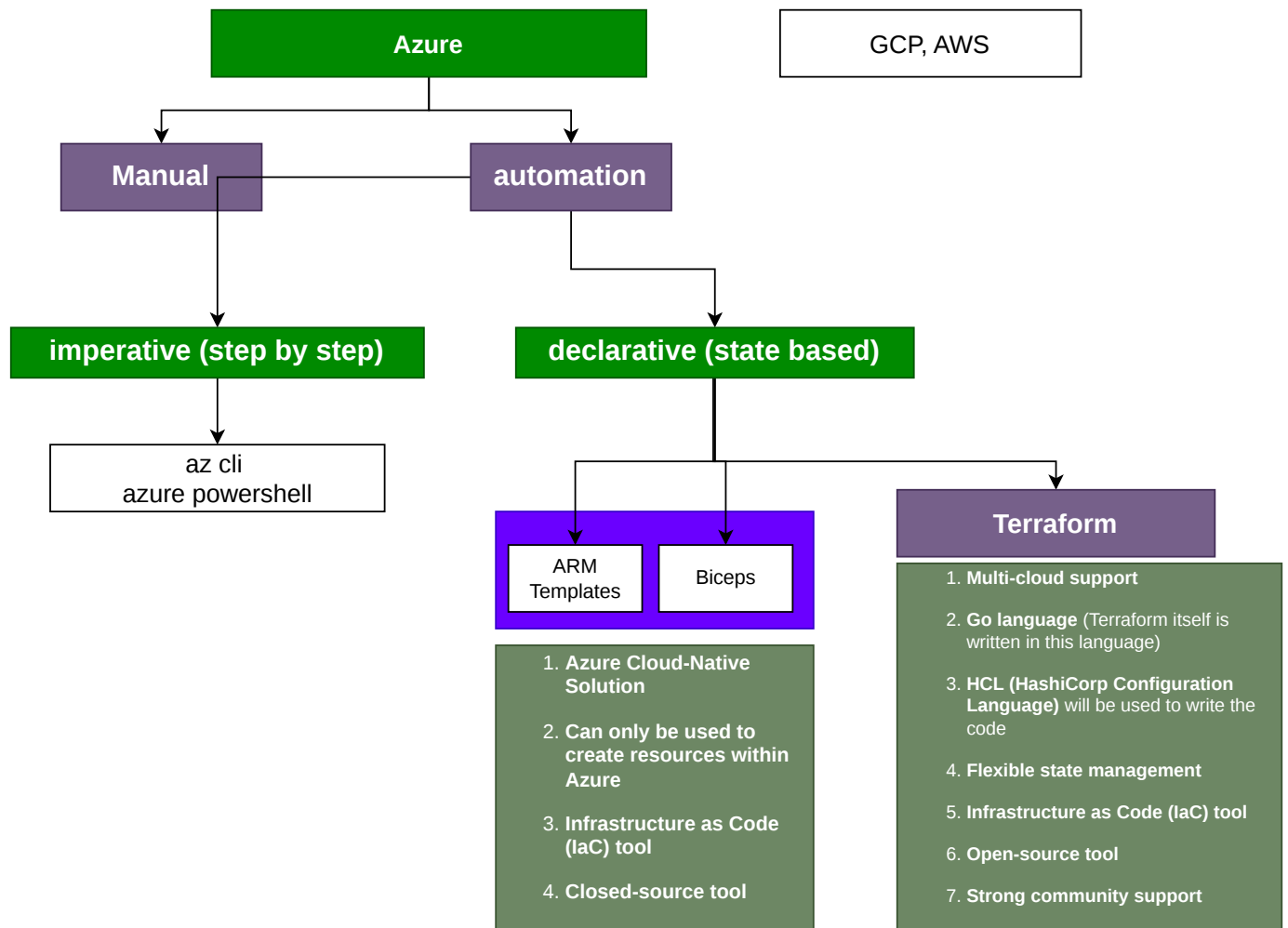


01-07-2026 (Terraform)

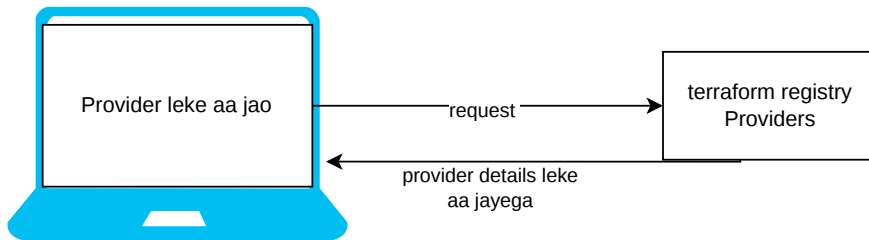


Terraform

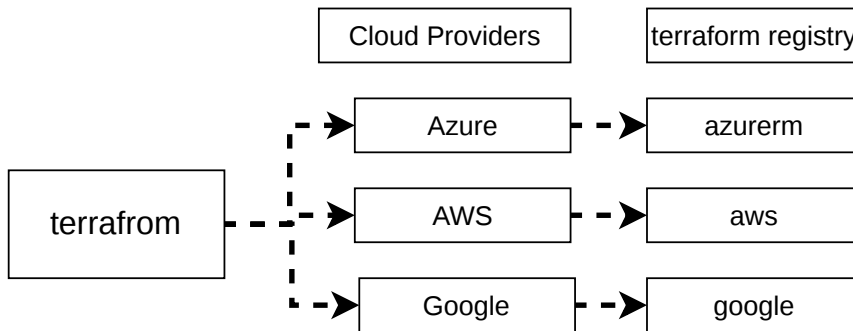
> This PC > Windows (C:) > terraform				
Sort View ...				
Name	Date modified	Type	Size	
LICENSE.txt	09-02-2026 01:09	Text Document	5 KB	
terraform.exe	22-06-2026 21:52	Application	1,16,939 KB	

Common Chiije hong

Providers: Azure, GCP, AWS.....6811 providers



terraform ko kisi specific cloud se baat karna sikhata hai



Azure login is not required to download the Terraform provider .exe file.

provider ko likha kesa jata hai ?



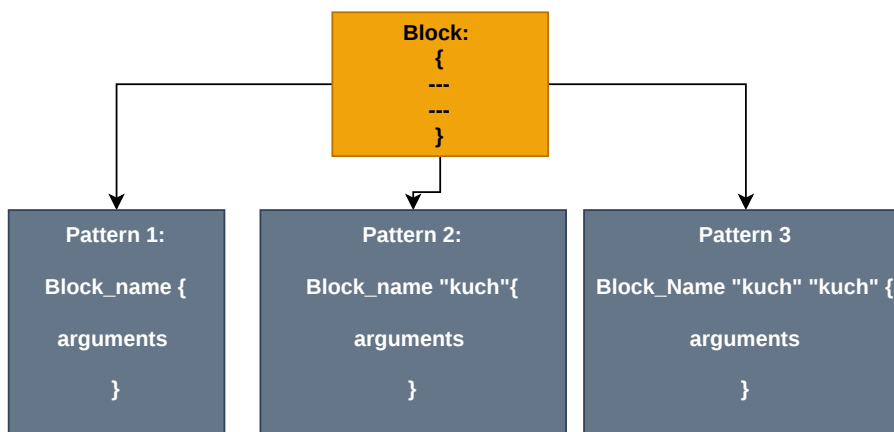
```
kuch_kuch1 kuch_kuch2 { kuch_hota_hai }
```

kuch_kuch1: resourec ka name aayega (block kis chij k hai wo identify krta hai)

```
terraform {
}
```

kuch_hota_hai : aruguments

```
{
  argument1
  argument2
}
```



Brackets name:

() : Parenthesis

[] : Square Bracket

{ } : curly Braces

Block Hai Kya?

```
block_name = {  
  arguments  
}
```

nhi ye block nhi hai

Block Hai Kya?

```
block_name : {  
  arguments  
}
```

nhi ye block nhi hai

```
E:\Dev\Devops\terraform>mkdir terraform-01-07-2026
```

```
E:\Dev\Devops\terraform>cd terraform-01-07-2026
```

```
E:\Dev\Devops\terraform\terraform-01-07-2026>code .
```

terraform file ki extension kya hoti hai: **.tf**

```
terraform {  
}
```

Block

```
required_providers {
```

Block

```
azurerm = {  
  source =  
  "hashicorp/azurerm"  
  version = "=4.1.0"
```

block nhi hai

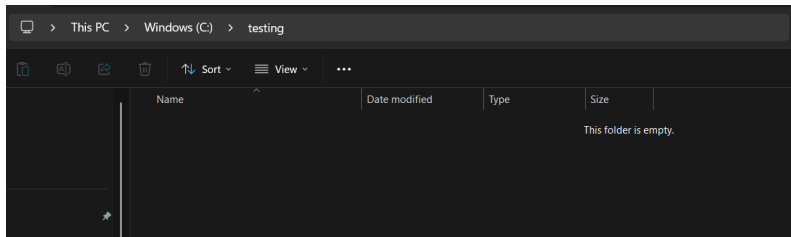
```
kuchbhi.tf > ...  
1 terraform {  
2   required_providers {  
3     azurerm = {  
4       source = "hashicorp/azurerm"  
5       version = "=4.1.0"  
6     }  
7   }  
8 }  
9
```

block k ander block ko
nested block bolter hai

```
3 azurerm = {  
4   source = "hashicorp/azurerm"  
5   version = "=4.1.0"  
6 }
```

Arguments

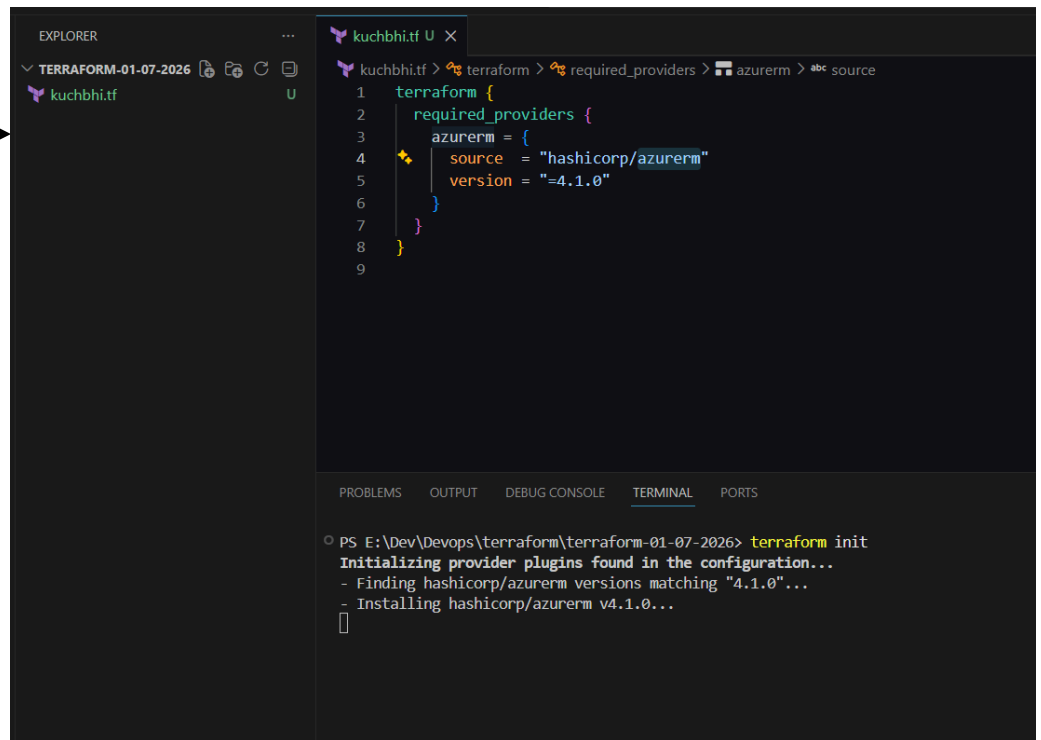
Terraform Commands



```
C:\testing>terraform init
Terraform initialized in an empty directory!

The directory has no Terraform configuration files. You may begin working
with Terraform immediately by creating Terraform configuration files.
```

terraform init



```
▼ TERRAFORM-01-07-2026
  ▼ .terraform\providers\registry.terraform.io\hashicorp\azurerm\4.79.0\windows_amd64
    📄 LICENSE.txt
    📄 terraform-provider-azurerm_v4.79.0_x5.exe
    📄 .terraform.lock.hcl
    🦋 kuchbhi.tf
```

2: terraform init -upgrade

.terraform.lock.hcl: ye lock kr deta hai version ko

agr hme koi or version use krna hai toh hme

terraform init -upgrade

```
PS E:\Dev\Devops\terraform\terraform-01-07-2026> terraform init
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/azurerm from the dependency lock file
```

Error: Failed to query available provider packages

Could not retrieve the list of available versions for provider hashicorp/azurerm: locked provider version 4.79.0 does not match configured version constraint 4.78.0; must use terraform init -upgrade to

To see which modules are currently depending on hashicorp/azurerm and what versions are specified, run terraform providers

```
PS E:\Dev\Devops\terraform\terraform-01-07-2026> terraform init -upgrade
```

The screenshot shows a VS Code editor with two Terraform configuration files open: `dusrasenario.tf` and `kuchbhi.tf`. The `EXPLORER` sidebar on the left shows the project structure for `TERRAFORM-01-07-2026`, including `.terraform\providers\registry.terraform.io\hashicorp\azurerm_v4.79.0_x5.exe`, `LICENSE.txt`, `terraform-provider-azurerm_v4.79.0_x5.exe`, `.terraform.lock.hcl`, `dusrasenario.tf`, and `kuchbhi.tf`.

The `dusrasenario.tf` file contains the following configuration:

```
1 terraform {
2   required_providers {
3     azurerm = {
4       source = "hashicorp/azurerm"
5       version = "=4.79.0"
6     }
7   }
8 }
```

The `kuchbhi.tf` file contains the following configuration:

```
1 terraform {
2   required_providers {
3     azurerm = {
4       source = "hashicorp/azurerm"
5       version = "=4.79.0"
6     }
7   }
8 }
```

The `TERMINAL` output shows the following error:

```
PS E:\Dev\Devops\terraform\terraform-01-07-2026> terraform init
Error: Terraform encountered problems during initialisation, including problems
with the configuration, described below.

The Terraform configuration must be valid before initialization so that
Terraform can determine which modules and providers need to be installed.

Error: Duplicate required providers configuration

on kuchbhi.tf line 2, in terraform:
2:   required_providers {

A module may have only one required providers configuration. The required providers were previously configured at dusrasenario.tf:2,5-23.
```

```
terraform fmt
```

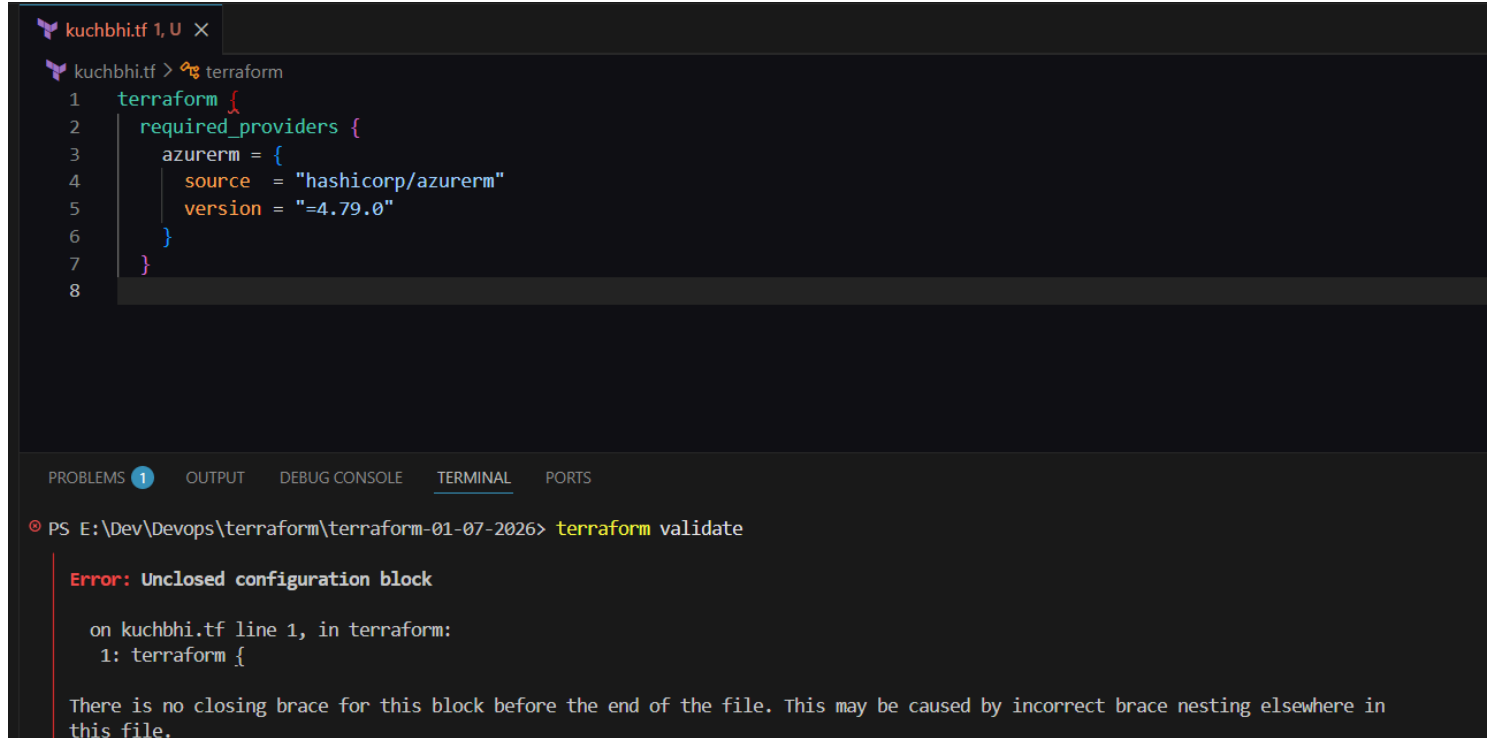
[illegible]

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

• PS E:\Dev\Devops\terraform\terraform-01-07-2026> terraform fmt
kuchbhi.tf
❖ PS E:\Dev\Devops\terraform\terraform-01-07-2026> 
```

```
kuchbhi.tf U X
kuchbhi.tf > terraform
1 terraform {
2     required_providers {
3         azurerm = {
4             source = "hashicorp/azurerm"
5             version = "=4.79.0"
6         }
7     }
8 }
```

terraform validate: Syntax Error ko check krta hai



The screenshot shows a VS Code editor with a file named `kuchbhi.tf` containing the following Terraform configuration:

```
1 terraform {  
2   required_providers {  
3     azurearm = {  
4       source  = "hashicorp/azurearm"  
5       version = "=4.79.0"  
6     }  
7   }  
8 }
```

The bottom panel shows the **TERMINAL** tab with the command `terraform validate` executed. The output indicates a syntax error:

```
PS E:\Dev\Devops\terraform\terraform-01-07-2026> terraform validate  
  
Error: Unclosed configuration block  
  
on kuchbhi.tf line 1, in terraform:  
1: terraform {  
  
There is no closing brace for this block before the end of the file. This may be caused by incorrect brace nesting elsewhere in this file.
```

```
10  
11 // Pattern 2  
12  
13 provider "azurearm" {  
14   features {}  
15 }  
16
```

terraform Comments

Terraform comments code ke andar likhe gaye wo simple notes hote hain jo run nahi hote, aur inka use sirf code ko samajhne ya explain karne ke liye kiya jata hai.

Type 1: `//`
Type 2: `#`

```
17  
18 // Pattern 3  
19  
20  
21 # resource "azurerm_resource_group" "resourceblockidentier1" {  
22 #   name      = "rg1"  
23 #   location = "Central US"  
24 # }  
25
```